Bridging Neuroscience and Artificial Intelligence

ITAI 4374

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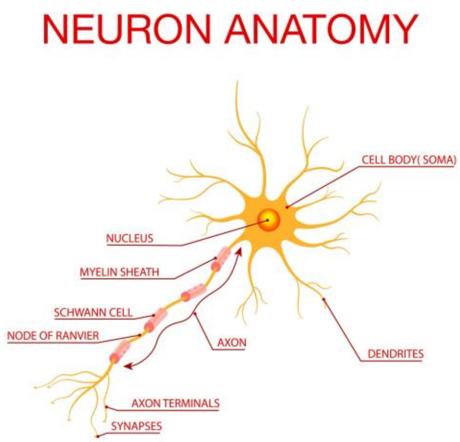
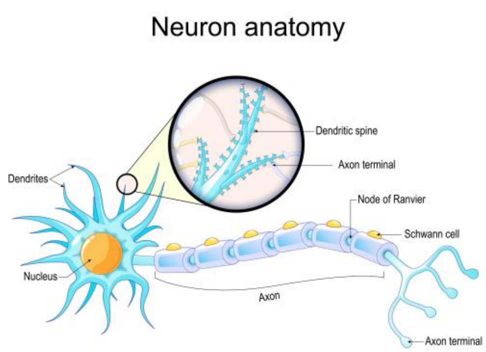
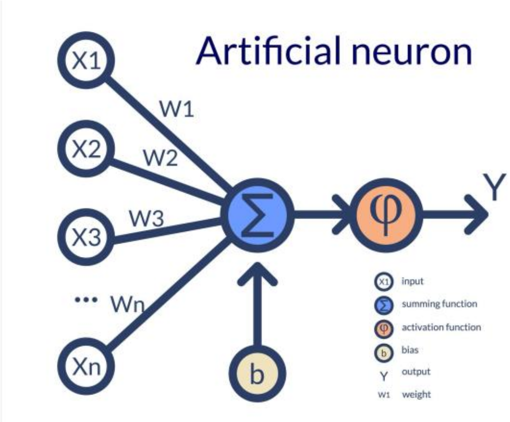
Biological vs Artificial Neurons



Neuron Anatomy 1



Neuron Anatomy 2



Artificial Neuron 1

Biological neurons are the fundamental processing units of the human brain, designed to transmit information through electrochemical signals (Squire et al., 1997). A biological neuron consists of dendrites that receive incoming signals, a cell body (soma) that integrates these signals, an axon that transmits outputs, and synapses that adjust signal strength based on experience. Artificial neurons are inspired by this structure but operate using mathematical abstractions rather than biological processes. Inputs are weighted, summed, passed through an activation function, and forwarded to the next layer (Heaton, 2017).

Biological inspiration strongly influenced early neural network design by introducing distributed processing, weighted connections, and learning through adaptation (Churchland & Sejnowski, 1992). However, artificial neurons are far simpler and lack the biochemical complexity, plasticity, and parallelism of biological systems (LeCun et al., 2015). Biological neurons are energy-efficient and adaptable but slow and noisy. Artificial neurons excel in speed, scalability, and precision but require massive datasets and computational resources. Both systems demonstrate strengths that continue to inform hybrid and biomimetic AI research (Heaton, 2017).

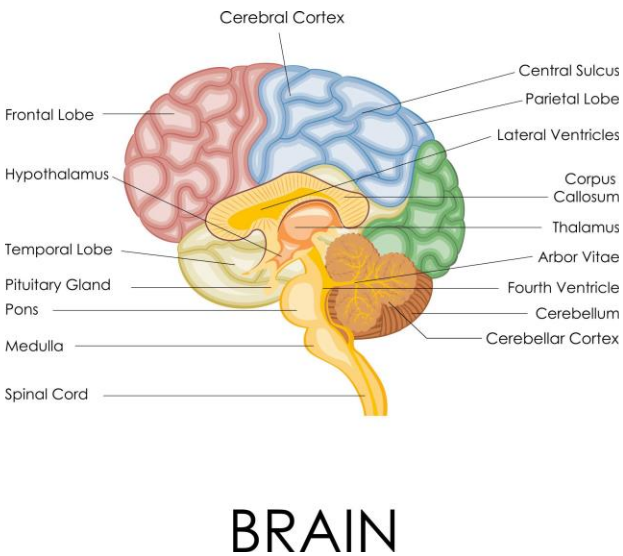
Comparison Table

Biological Component	**Function**	**Artificial Equivalent**	**AI Example**
Dendrites	Receive signals	Input features	CNN input pixels
Soma	Integrates signals	Summation + activation	Dense layer neuron
Axon	Sends output	Output value	Layer output
Synapse	Adjusts signal strength	Weight parameter	Learned model weights

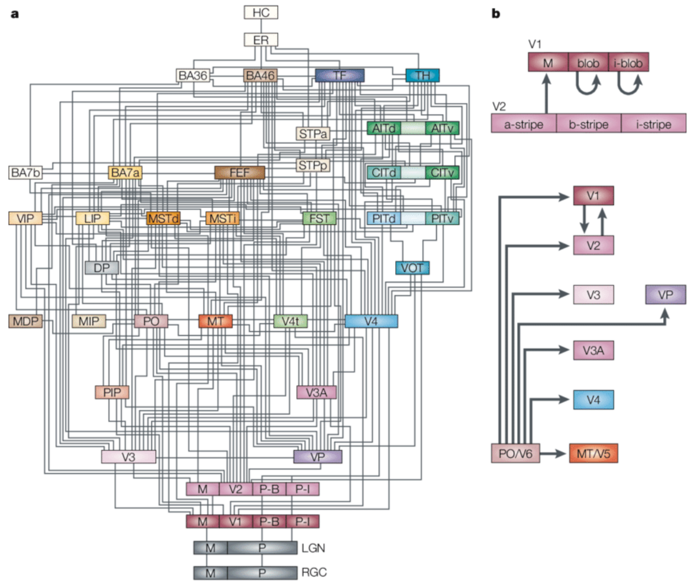
Summarized Table 1

Brain Architecture and AI Design

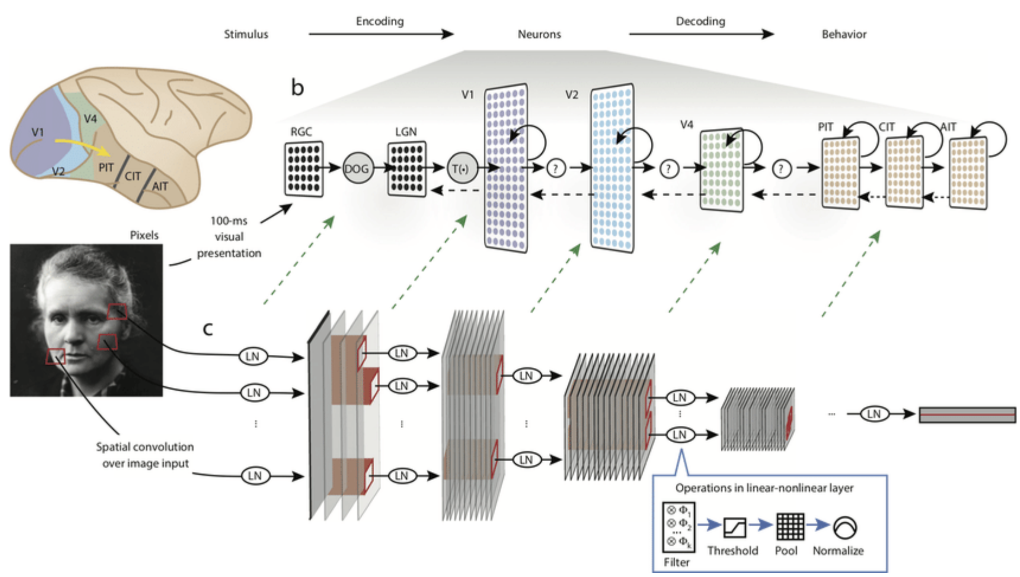
Hierarchical Organization of the Brain



Brain Architecture 1



AI Architecture 1

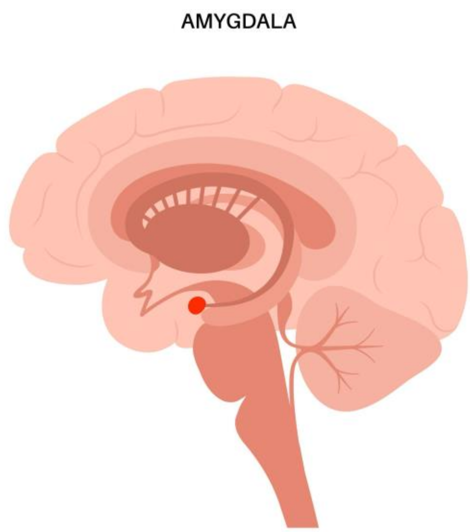


Comparison of Brian and AI Architecture 1

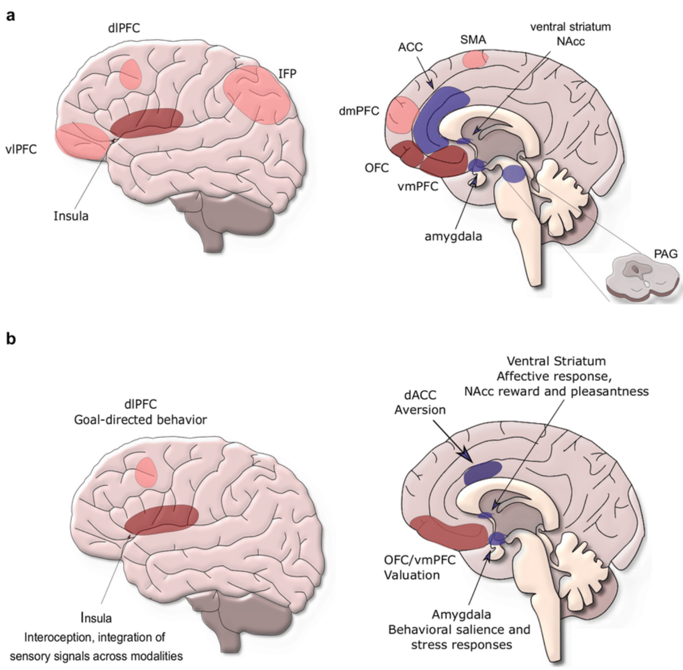
The human brain is hierarchically organized, with lower-level regions handling basic sensory input and higher-level regions managing complex cognition (Squire et al., 1997). The visual cortex processes visual information in stages—detecting edges, shapes, and objects—similar to how convolutional neural networks (CNNs) extract features layer by layer. Early CNN layers identify simple patterns, while deeper layers recognize complex objects (LeCun et al., 2015).

Recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) models mirror functions of the hippocampus by retaining sequential and temporal information (Squire et al., 1997). Attention mechanisms resemble biological attention systems by prioritizing relevant stimuli while suppressing irrelevant inputs, improving efficiency and focus (Heaton, 2017).

Amygdala and Emotion Processing



Amygdala 1



Structure of the Brain 1

The amygdala plays a central role in emotional processing, especially fear, risk evaluation, and rapid decision-making (Squire et al., 1997). Insights from biological emotion processing can inform AI systems by enabling context-aware risk assessment, emotionally sensitive pattern recognition, and adaptive learning strategies (Churchland & Sejnowski, 1992). Emotion-inspired AI could improve human-AI interaction, safety-critical decision systems, and autonomous agents operating in uncertain environments (LeCun et al., 2015).

Integration Analysis and Future Implications

Understanding biological neural networks offers valuable guidance for improving artificial intelligence design. Biological systems demonstrate remarkable efficiency, adaptability, and robustness despite operating with limited energy and noisy signals. Future AI architectures could incorporate principles such as sparse connectivity, continuous learning, and local adaptation inspired by synaptic plasticity. Neuromorphic computing and spiking neural networks already explore biologically inspired timing-based information processing to reduce energy consumption and improve real-time learning (Heaton, 2017).

Biological networks outperform artificial systems in generalization and learning from small datasets, whereas artificial networks excel in speed, accuracy, and scalability. While modern AI models require large datasets and retraining, biological systems learn incrementally and transfer knowledge across domains. Mimicking these properties could address current challenges such as catastrophic forgetting and poor adaptability (LeCun et al., 2015).

From an efficiency standpoint, the human brain consumes roughly 20 watts of power, while large AI models require vast computational resources. Integrating biological strategies such as parallel distributed processing and event-driven computation may significantly reduce AI's energy footprint (Churchland & Sejnowski, 1992).

Future research directions include hybrid systems that combine biological learning principles with artificial computation, brain-computer interfaces, and bio-inspired architectures capable of lifelong learning. As neuroscience continues to reveal how cognition, emotion, and perception interact, AI systems can evolve beyond static pattern recognition toward adaptive, resilient, and context-aware intelligence that more closely mirrors human cognition.

References

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